Junior Division

Questions 1 to 10, 3 marks each
1-10 题，每题 3 分

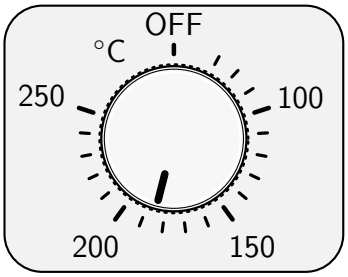
1. The metal in an Australian 20 cent coin is one-quarter nickel, and the rest is copper. What fraction is copper?

(A) $\frac{3}{4}$ (B) $\frac{1}{2}$ (C) $\frac{1}{4}$ (D) $\frac{4}{4}$ (E) 0

The fraction that is copper is $1 - \frac{1}{4} = \frac{3}{4}$,

hence (A).

2. Owen set the temperature dial on his oven as shown. What temperature is it set to?



(A) 175 °C (B) 185 °C (C) 195 °C (D) 205 °C (E) 215 °C

Between 150 and 200 there are 5 intervals, so each interval represents 10 °C. The dial is $3\frac{1}{2}$ intervals above 150 °C which is therefore 185 °C,

hence (B).

3. If the day before yesterday was Monday, then the day after tomorrow will be

(A) Monday (B) Tuesday (C) Wednesday (D) Thursday (E) Friday

If the day before yesterday was Monday, then yesterday was Tuesday and today must be Wednesday. So tomorrow is Thursday and the day after tomorrow is Friday,

hence (E).

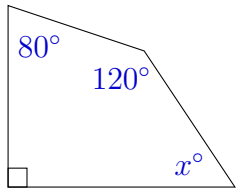
4. The value of $\frac{2024}{2+0+2+4}$ is

(A) 1 (B) 126.5 (C) 253 (D) 506 (E) 1012

Evaluating, $\frac{2024}{2+0+2+4} = \frac{2024}{8} = 253$,

hence (C).

5. What is the value of x in the diagram?



(A) 40 (B) 50 (C) 60 (D) 70 (E) 80

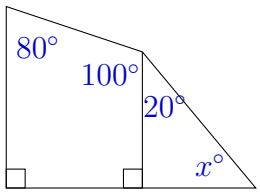
Alternative 1

The angles in a quadrilateral add to 360°, so $x = 360 - 90 - 80 - 120 = 70$,

hence (D).

Alternative 2

With another vertical line added, the cointerior angle of 100° can be found, and hence the 20° angle shown.

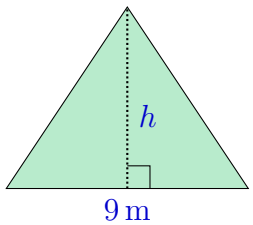


Finally $x° = 180° - 90° - 20° = 70°$,

hence (D).

---

6. The area of the triangle shown is $36\,\text{m}^2$. What is the length h?



(A) 2 m (B) 4 m (C) 4.5 m (D) 8 m (E) 16 m

Using the formula for the area of a triangle $A = \frac{1}{2}bh$,

$$\begin{aligned} \tfrac{1}{2}bh &= A \\ \tfrac{1}{2} \times 9 \times h &= 36 \\ 9 \times h &= 72 \\ h &= 8 \end{aligned}$$

hence (D).

---

7. What is the difference between $3 \times (4+2)$ and $(3 \times 4) + 2$?

(A) 0 (B) 2 (C) 4 (D) 6 (E) 8

Evaluating, $3 \times (4+2) = 3 \times 6 = 18$ and $(3 \times 4) + 2 = 12 + 2 = 14$. Then $18 - 14 = 4$, hence (C).

---

8. What number must be divided by 100 to get 0.2?

(A) 0.002 (B) 0.02 (C) 0.2 (D) 2 (E) 20

Let x be the number. Then $x \div 100 = 0.2$ so $x = 100 \times 0.2 = 20$, hence (E).

---

9. The average of 2024 and an unknown number is $2\frac{1}{2}$. What is the unknown number?

(A) 2019 (B) -2019 (C) 2029 (D) -2029 (E) -2024

The two numbers add to $2 \times (2\frac{1}{2}) = 5$ so the unknown number is $5 - 2024 = -2019$, hence (B).

---

10. The *Balga* is a grass tree found in south-western Australia. Balgas grow between 1 and 2 centimetres taller each year. Approximately how old is a Balga that is 4 metres tall?

(A) 10 years (B) 30 years (C) 60 years (D) 300 years (E) 1000 years

Each 100 years, a Balga will grow between 1 and 2 metres. To reach 4 metres will take between 4 centuries, at 1 metre per century, or 2 centuries, at 2 metres per century. Of the answers given, 300 years is the best approximation, hence (D).

Questions 11 to 20, 4 marks each
11-20 题，每题 4 分

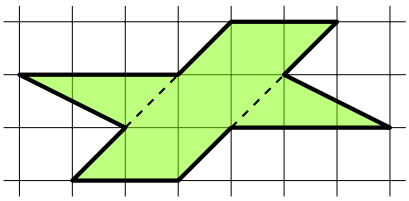
11. Danilo's pasta recipe requires one 60-gram egg for each 100 grams of flour, and no other ingredients. To make 800 grams of pasta, how many eggs does he need?

(A) 5 (B) 6 (C) 7 (D) 8 (E) 9

Suppose that one serve of pasta requires one 60 gram egg and 100 grams of flour, which is a total of 160 grams of pasta. Then 800 grams of pasta amounts to $800 \div 160 = 5$ serves. Since each serve requires one egg, 5 eggs are needed,

hence (A).

12. What is the area of the shape drawn in the grid?



(A) 7 (B) 8 (C) 9 (D) 10 (E) 11

Alternative 1

Counting grid squares, the area is $2 \times 3 + 1 \times 3 = 9$,

hence (C).

Alternative 2

The central parallelogram has area $bh = 2 \times 3 = 6$. Each triangle has area $\frac{1}{2}bh = \frac{1}{2} \times 3 \times 1 = \frac{3}{2}$. The total shaded area is $6 + 2 \times \frac{3}{2} = 9$,

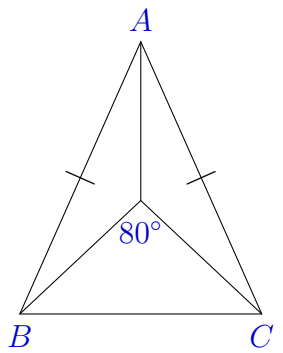
hence (C).

---

13. The isosceles triangle ABC is divided into three smaller isosceles triangles as shown.

What is the value, in degrees, of $\angle ABC$?

(A) 50 (B) 55 (C) 60 (D) 65 (E) 70



Alternative 1

The unlabelled angles in the lower isosceles triangle are each $\dfrac{180-80}{2} = 50$ degrees. The obtuse angles in the upper two triangles are each $\dfrac{360-80}{2} = 140$ degrees, so the other angles of these two triangles are each $\dfrac{180-140}{2} = 20$ degrees. Combining these, $\angle ABC = 50 + 20 = 70$ degrees,

hence (E).

Alternative 2

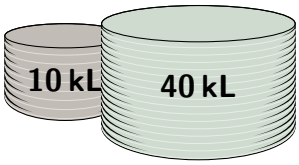
As in Alternative 1, the acute angles of the upper two isosceles triangles are each $20°$. So $\angle BAC = 20° + 20° = 40°$. Angles ABC and ACB are both $\frac{180-40}{2} = 70°$,

hence (E).

---

14. We have two water tanks for bushfire emergencies. The 10 kL tank is 80% full and the 40 kL tank is 30% full. Regulations state that we must have at least 30 kL available.

How much more water do we need to add to our tanks?



(A) 0 kL (B) 5 kL (C) 10 kL (D) 15 kL (E) 20 kL

The amount of water currently stored is $\frac{80}{100} \times 10 + \frac{30}{100} \times 40 = 8 + 12 = 20$ kilolitres. So another 10 kL is required to make 30 kL,

hence (C).

---

15. I have a pad of 100 sticky notes. Each note is a 75 mm × 75 mm square.

Using most of the pad, I completely cover all the faces of a cube. None of the notes folds over an edge of the cube.

What is the largest the side length of the cube could be?

(A) 22.5 cm (B) 30 cm (C) 45 cm (D) 60 cm (E) 75 cm

Since $100 \div 6$ is between 16 and 17, one possibility is to place 16 sticky notes on every face in a 4×4 grid of notes. This can cover a cube of side length 300 mm = 30 cm.

No more is possible, since at least one face will have 16 or fewer sticky notes, and these can, at most, exactly cover one face of a 300 mm cube,

hence (B).

---

16. In this square, all rows and columns have the same sum.
Find the average of A, B, C and D.

A	11	B
X	5	Y
D	7	C

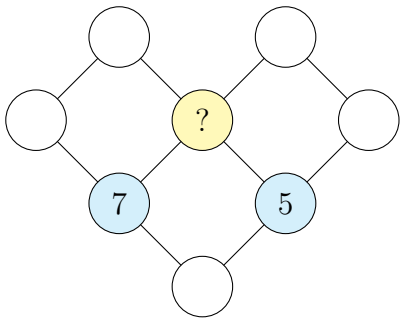
(A) 7 (B) 8 (C) 9 (D) 10 (E) 11

The second column adds to $11 + 5 + 7 = 23$ so this is the sum of each row and each column. Then $A + 11 + B = 23$ so that $A + B = 12$. Similarly, $X + Y = 18$ and $C + D = 16$. (For instance, $A = B = 6$, $C = D = 8$ and $X = Y = 9$ is one way to complete the grid.) Then the average of A, B, C and D is
$(A + B + C + D) \div 4 = 28 \div 4 = 7$,

hence (A).

---

17. In this puzzle, the numbers from 1 to 8 are to be placed in the 8 circles. For each of the 3 squares, the sum of the 4 numbers on the corners must be 20.
With 5 and 7 placed as in the diagram, what number goes in the middle circle?

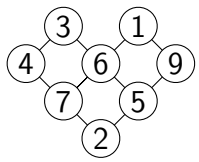


(A) 1 (B) 2 (C) 3 (D) 4 (E) 6

Alternative 1
In the bottom square, we know $5 + 7 = 12$ so the other two numbers add to $20 - 12 = 8$. Since 5 and 7 are already used, these two numbers must be 2 and 6, and there are two cases — the middle number is either 2 or 6.

If the middle number was 2, then the right square has $2 + 5 = 7$ so the other two numbers add to $20 - 7 = 13$. However, 13 can only be made as $5 + 8$ or $6 + 7$ and 5 and 7 are already used. So the middle number can't be 2.

If the middle number is 6, the bottom number is 2, and we can place 3 and 4 in the top left and 1 and 9 in the top right. This is the only solution, so the middle number must be 6. One possibility for the complete puzzle is shown,



hence (E).

Alternative 2

Try placing 1 in the middle, so that $2+3+4+6+8=23$ is unused. However, we have targets of 12 on the left, 14 on the right and 7 at the bottom, for a total of 33.

With 2 in the middle, $1+3+4+6+8=22$ is unused and the targets are $11+13+6=30$.

Every extra 1 in the middle decreases the unused total by 1 and decreases the total target by 3. This narrows the gap by 2. They are both equal to 18 when the middle number is 6. With 6 in the middle, the rest of the grid is easily filled in, as in the earlier solution,

hence (E).

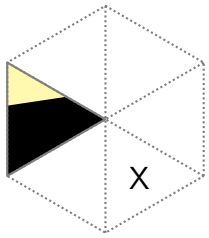
Alternative 3

The 8 numbers add to 36, but the 3 squares add to 60. The extra 24 is from counting the numbers 5 and 7 a second time and from counting the middle number 3 times. So twice the middle number is $24-5-7=12$ and the middle number is 6,

hence (E).

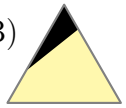
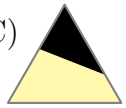
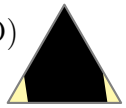
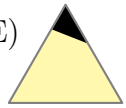
---

18. A regular hexagon made from yellow cardboard has a thick black stripe painted across it. The hexagon is cut up into six equilateral triangles and the pieces are mixed up. This diagram shows one of the pieces in its original position.



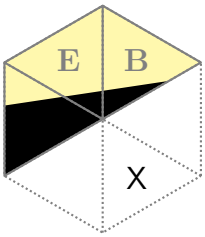
The other five pieces are shown in options (A) to (E).

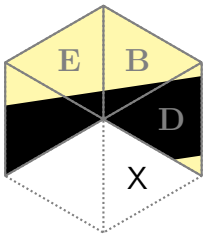
Which piece was in the position marked with the X?

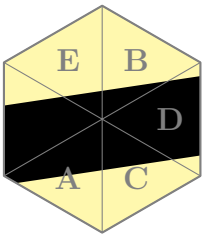
(A)
(B) 
(C) 
(D) 
(E) 

Alternative 1

Continuing the top edge of the black stripe to the right as shown in the first diagram, the top two pieces must be (E) and (B). Only piece (D) has a small yellow corner, which matches the top of the stripe as shown in the second diagram. The third diagram shows where the remaining pieces fit.







,

hence (C).

Alternative 2

The lower-left triangle has a fully black edge, so must be (A). The lower-right triangle has an edge that is slightly less than half black, which can only be (C). So triangle (C) goes in position X,

hence (C).

---

19. A three-digit number has all digits nonzero. These digits are multiplied together and then this product is subtracted from the original number. What is the largest possible value of this difference?

(A) 890 (B) 901 (C) 909 (D) 910 (E) 990

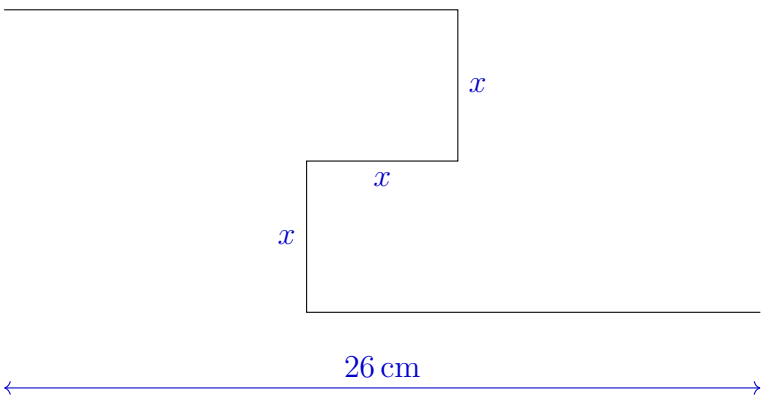
Let N be the three-digit number, P the product of its digits and $D = N - P$. Examples like $N = 911$ show that the biggest possible D is between 900 and 999, so both D and N will begin with 9. Write $N = 900 + 10b + c$ where b and c are nonzero digits. Then $P = 9bc$ and $D = 900 + 10b + c - 9bc = 900 + 10b - c(9b - 1)$.

For any nonzero b, $9b - 1 > 0$ so that D is greatest when $c = 1$. Then $D = 901 + b$ which has maximum value 910 when $b = 9$ and $n = 991$,

hence (D).

---

20. A piece of wire of length 50 cm is bent into the shape shown, where each bend in the wire forms a right angle. The three internal sections of the bent wire are each the same length, x cm. What is the value of x?



(A) 4 (B) 6 (C) 8 (D) 10 (E) 12

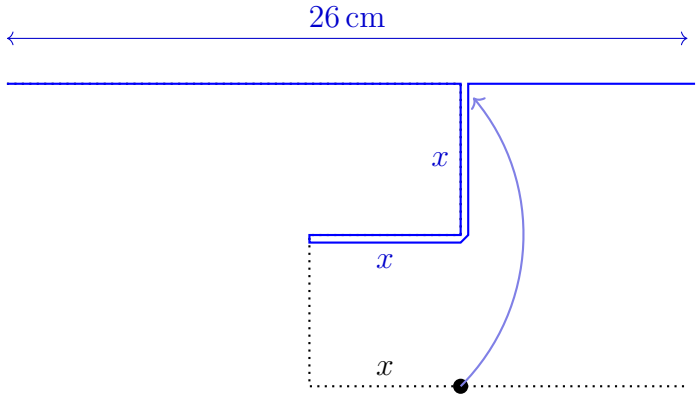
Alternative 1

The length of the sum of the two end sections of the bent wire is $26 + x$ because the bottom section starts x back from the top section. Hence the total length of the bent wire is $26 + 4x = 50$ giving $4x = 24$ so $x = 6$,

hence (B).

Alternative 2

Bend the wire further as indicated by the dotted lines.

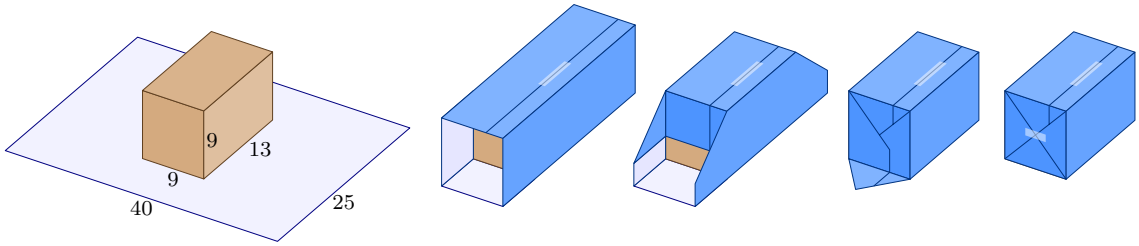


The doubled-up part of the wire is $50 - 26 = 24\,\text{cm}$, so that $x = 24 \div 4 = 6\,\text{cm}$,

hence (B).

Questions 21 to 25, 5 marks each
21-25 题，每题 5 分

21. Amisha is wrapping a present for her little brother. The present is in a box measuring $9\,\text{cm} \times 9\,\text{cm} \times 13\,\text{cm}$ and the sheet of wrapping paper is $40\,\text{cm} \times 25\,\text{cm}$. She wraps the present as tightly as possible, as shown.



What percentage of the wrapping paper is no longer visible from any angle?

(A) 25% (B) 30% (C) 33% (D) 37% (E) 42%

The area of wrapping paper that *is* visible from the outside is the same as the total area of the six faces of the box. This is the *total surface area* of the box. There are two square faces with area $9 \times 9 = 81\,\text{cm}^2$ and four rectangular faces with area $9 \times 13 = 117\,\text{cm}^2$. So this total area is

$$2 \times 81 + 4 \times 117 = 630\,\text{cm}^2$$

The full area of the wrapping paper is $40 \times 25 = 1000\,\text{cm}^2$. Therefore the percentage of wrapping paper which is visible is

$$\frac{630}{1000} \times 100\% = 63\%$$

The remaining 37% is not visible,

hence (D).

22. A long line of ping-pong balls numbered 1, 2, 3, ... is sitting in order in a tilted chute, as shown below.

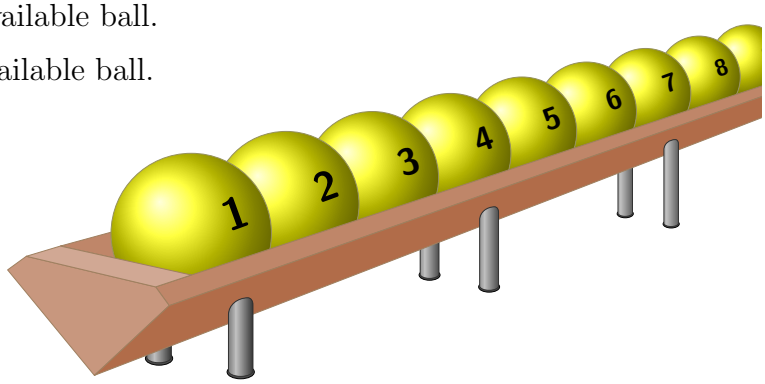
Xavier, Yasmin and Zebedee take turns removing balls from the chute. When a ball is removed, the rest roll down to fill the gap. When Zebedee has taken his ball, it is Xavier's turn again and they continue in the same order for as long as there are balls remaining.

Xavier always takes the second available ball.

Yasmin always takes the third available ball.

Zebedee always takes the first available ball.

Which statement is **true**?



(A) Xavier takes ball 48.

(B) Yasmin takes ball 49.

(C) Zebedee takes ball 50.

(D) Yasmin takes ball 51.

(E) Xavier takes ball 52.

Xavier takes the second ball, which is number 2. Yasmin then takes the third ball from the remaining list 1, 3, 4, ..., which is number 4. Zebedee then takes the first ball, which is number 1. This leaves the list 3, 5, 6, ...

We can tabulate the choices as shown below, crossing off each number in the first row as it is removed. After the first four rounds we have the following:

List	~~1~~	~~2~~	~~3~~	~~4~~	~~5~~	~~6~~	~~7~~	~~8~~	~~9~~	~~10~~	~~11~~	12	~~13~~	14	...
X		2			5			8			11				...
Y				4			7			10			13		...
Z	1		3			6			9						...

With the exception of Zebedee taking 1 and then 3 in the first two rounds, the pattern of taking every third number continues for all players. In particular, Zebedee takes all multiples of 3, so we can predict that he will take 48 and the table will continue as follows:

List	...	~~48~~	~~49~~	~~50~~	~~51~~	~~52~~	~~53~~	...
X				50			53	
Y			49			52		
Z		48			51			

So the only true statement is that Yasmin takes ball 49,

hence (B).

23. To make the stamps easier to pull apart, sheets of postage stamps have rows of holes punched into them.

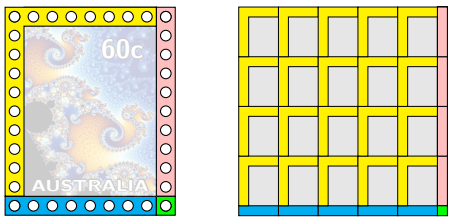
In the sheet shown, each stamp has 7 holes along the top and bottom edges, 9 holes along the sides and 1 more at each corner. Holes are included around all four edges of the sheet so the borders can be torn off as well.

The full sheet has 20 stamps arranged in 4 rows of 5. How many holes are there altogether?

(A) 320 (B) 391 (C) 421 (D) 451 (E) 640

Alternative 1

Break up the perforations for each stamp into four parts as shown on the left: the yellow hook shape of $7 + 1 + 9 = 17$ holes, the blue base strip of $1 + 7 = 8$ holes, the pink side strip of $1 + 9 = 10$ holes and the green corner with 1 hole.



To avoid double counting overlapping edges, the entire sheet can be accounted for by the arrangement shown on the right: 20 yellow hooks (one per stamp), 5 blue base strips, 4 pink side strips and 1 green corner.

So the total number of holes is

$$(20 \times 17) + (5 \times 8) + (4 \times 10) + 1 = 421,$$

hence (C).

Alternative 2

There are 6 columns of holes and 5 rows of holes, each with 41 holes. Counting this as $41 \times 11 = 451$, we have double-counted $6 \times 5 = 30$ corner holes that are in both a row and a column. Subtracting to compensate for double-counting, there are $451 - 30 = 421$ holes, hence (C).

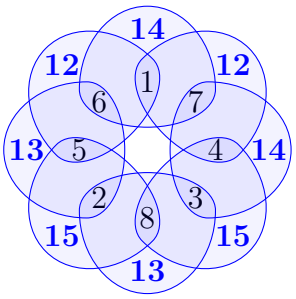
24. Sally arranges the numbers from 1 to 8 in a circle and then calculates the sum of each set of 3 adjacent numbers. She would like an arrangement in which the smallest of these sums is as large as possible. If she chooses such an arrangement, what will this smallest sum be?

(A) 10 (B) 11 (C) 12 (D) 13 (E) 14

Let the smallest of the sums of three adjacent numbers be x.

Then adding from the number 1 around the circle and back to 1 adds 9 numbers in a row, so this value is at least $3x$. However these 9 numbers add to $1 + 1 + 2 + 3 + \cdots + 8 = 37$. Hence $37 \geqslant 3x$ so that $12\frac{1}{3} \geqslant x$ and since x is a whole number, $x \leqslant 12$.

Once it is known that 12 has a chance of being the largest minimum, it is straightforward to find one of the many arrangements where all sums of 3 adjacent numbers (shown in blue) are 12 or more, such as the arrangement shown. This confirms that the smallest possible sum of 3 adjacent numbers is 12,



hence (C).

---

25. A *totally even number* is a positive integer in which all the digits are even. The list of all totally even numbers begins as follows:

$$2, 4, 6, 8, 20, 22, 24, 26, 28, 40, \ldots$$

The number 40 is in the 10th position in this list. In which position is the totally even number 2024?

(A) 57 (B) 132 (C) 257 (D) 512 (E) 1012

The units digit cycles though the 5 even digits: $2, 4, 6, 8, 0, 2, 4, 6, \ldots$. To make the cycle clearer, temporarily start with the number 0, so that the cycles finishes after the 5th, 10th, … position, and the tens digit increases there as well.

Every 5 positions, the tens digit increases in the same cycle of even digits, taking 25 positions to cycle. Likewise the hundreds digit increases every 25 positions, for a cycle of 125 positions and the thousands digit increases every 125 positions. So the list up to 2024 is

$$\underbrace{0, 2, 4, \ldots, 888}_{125}, \underbrace{2000, 2002, \ldots, 2008}_{5}, \underbrace{2020, 2022, 2024}_{3}$$

which has 2024 in position 133. Without the initial 0, 2024 is in position 132,

hence (B).

For questions 26 to 30, shade the answer as an integer from 0 to 999 in the space provided on the answer sheet.

问题 26-30 的答案为 0-999 之间的整数，请将答案填涂在答题卡对应区域。

Questions 26–30 are worth 6, 7, 8, 9 and 10 marks, respectively.

第 26-30 题分别为 6, 7, 8, 9, 10 分。

26. Ringo has 5 cards with the numbers 2, 3, 4, 5 and 6 on them. He wrote a list of all possible two-digit numbers that can be made with these cards.

Find the sum of all numbers in his list that are not prime.

Alternative 1

We first consider the sum of all possible numbers, including primes. The numbers in the list are

$$23, 24, 25, 26, 32, 34, 35, 36, 42, 43, 45, 46, 52, 53, 54, 56, 62, 63, 64, 65$$

Each digit occurs in the units position 4 times, so the units digits add to $4 \times (2+3+4+5+6) = 80$. Similarly the tens digits add to 80, so that the total of all 20 numbers is $80 + 800 = 880$.

The only primes in the list are 23, 43 and 53, which add to 119. So the required sum is $880 - 119 = 761$,

hence (761).

Alternative 2

There are $5 \times 4 = 20$ choices of tens digit then ones digit. These 20 numbers can be arranged into 10 pairs, each adding to 88:

$$88 = 23 + 65 = 24 + 64 = \cdots$$

These 10 pairs add to $10 \times 88 = 880$.

As in Alternative 1, subtracting the three primes gives $880 - 119 = 761$,

hence (761).

27. A farmer, Jenni, grows oranges and grapefruit. At her last harvest she picked 120 grapefruit and a number of oranges. However, one-tenth of all grapefruit look like oranges and one-tenth of all oranges look like grapefruit. Overall, one-fifth of all fruits look like grapefruit.

How many oranges did Jenni harvest?

Of the 120 grapefruit she picks, 12 look like oranges, so 108 look like grapefruit.

Suppose Jenni picks $10x$ oranges. Then $9x$ look like oranges and x look like grapefruit.

Of the $120 + 10x$ fruit picked, one fifth look like grapefruit, which is $24 + 2x$. This is also equal to $108 + x$. Solving,

$$24 + 2x = 108 + x$$
$$x = 108 - 24 = 84$$

So the number of oranges picked is $10x = 840$,

hence (840).

---

28. A rectangular piece of paper would lose $999\,\text{cm}^2$ of area if a 3 cm wide border was trimmed from all four sides. If instead the trimmed border was 2 cm wide, what area, in square centimetres, would be lost?

Alternative 1

Let the dimensions of the rectangle be x cm $\times$ y cm. When the 3 cm border is trimmed, the width and height both reduce by 6 cm. So the area lost, in square centimetres, is

$$xy - (x-6)(y-6) = xy - (xy - 6x - 6y + 36) = 6(x+y) - 36.$$

Setting this area equal to 999 and solving shows that $x + y = \frac{1035}{6} = \frac{345}{2}$.

If the border is 2 cm wide instead, the width and height both reduce by 2 cm. So the area lost, in square centimetres, is

$$xy - (x-4)(y-4) = 4(x+y) - 16 = 4 \times \frac{345}{2} - 16 = 2 \times 345 - 16 = 674$$

hence (674).

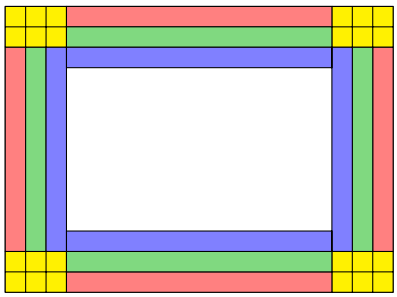
Alternative 2

Trim the border 1 cm at a time, creating a sequence of border trimmings. Each can be made up of 4 rectangles, each 1 cm wide, and each rectangle loses 2 cm length from one to the next. So the borders' areas are 3 numbers in arithmetic progression with difference -8. Since they sum to 999, their average is 333, which is the middle number.

Consequently, the borders are 341, 333 and 325 cm^2. If only the outer two are removed, this would be $341 + 333 = 674\,\text{cm}^2$,

hence (674).

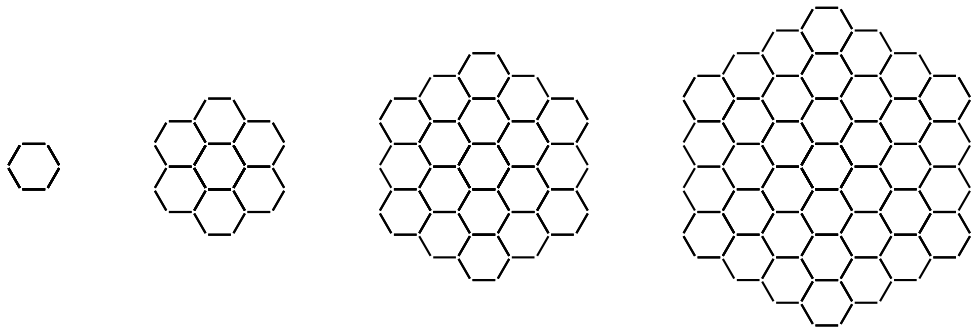
Alternative 3



$$\bigcirc + \bigcirc + \bigcirc + \bigcirc = 999\,\text{cm}^2$$
$$\bigcirc = 24\,\text{cm}^2$$
$$\bigcirc + \bigcirc + \bigcirc = 999 - 24 = 975\,\text{cm}^2$$
$$\bigcirc = \bigcirc = \bigcirc = 975 \div 3 = 325\,\text{cm}^2$$
$$\bigcirc + \bigcirc + \bigcirc = 999 - 325 = 674\,\text{cm}^2$$

hence (674).

29. Rodney builds increasingly large shapes out of unit rods. The smallest uses six rods and each larger shape adds a layer of hexagons around the previous shape.



Starting with 2000 rods, Rodney builds a single shape like this, as large as possible. How many rods will he have left over?

Alternative 1

Suppose the shape has N hexagons on each side. Consider first the horizontal rods, counted in vertical columns. These columns go

$$\underbrace{(N+1)+(N+2)+\cdots+(2N-1)}_{N-1}+(2N)+\underbrace{(2N-1)+\cdots+(N+1))}_{N-1}$$

which we rearrange by making pairs that add to $3N$, like $(N+1)+(2N-1)=3N$, $(N+2)+(2N-2)=3N$ and so on, giving that the number of horizontal rods is

$$\underbrace{3N+3N+\cdots+3N}_{N-1}+(2N)=(N-1)(3N)+2N=3N^2-N$$

There are two other directions, but these have the same number of rods, making a total of $9N^2-3N$ rods. Estimating, $2000\approx 9N^2-3N\approx 9N^2$ so that $N\approx\sqrt{2000\div 9}\approx\sqrt{225}=15$. So we test N near 15:

N	14	15	16
$9N^2-3N$	1722	1980	2256

So the greatest N could be is 15, and then there are $2000-1980=20$ rods left,

hence (20).

Alternative 2

In shape number n, let $R\approx 2000$ be the number of rods and H the number of hexagons. There are $B=6(2n-1)=12n-6$ rods around the boundary and $R-B$ non-boundary rods. Counting 6 rods around each hexagon counts every boundary rod once and every non-boundary rod twice. So $6H=B+2(R-B)=2R-B=2R-12n+6$ and then $R=3H+6n-3$.

By considering concentric layers of hexagons,

$$\begin{aligned} H &= 1 + 6 + 12 + \cdots + 6(n-1) \\ &= 1 + 6 \times \big(1 + 2 + \cdots + (n-1)\big) \\ &= 1 + 6 \times \tfrac{n(n-1)}{2} \\ &= 3n^2 - 3n + 1 \\ \Rightarrow R = 3H + 6n - 3 &= 3(3n^2 - 3n + 1) + 6n - 3 \\ &= 9n^2 - 3n \end{aligned}$$

Then $9n^2 - 3n - R = 0$ where $R \leqslant 2000$. We solve this quadratic for $n > 0$:

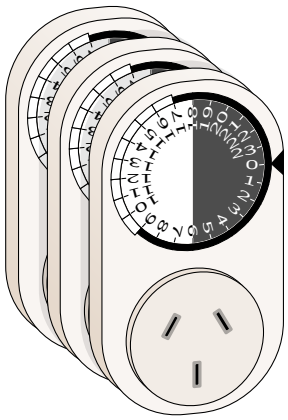
$$\begin{aligned} 9n^2 - 3n - R = 0 \\ n &= \frac{1}{18}\Big(3 + \sqrt{9 + 36R}\Big) \\ &\leqslant \frac{1}{6} + \frac{1}{18}\sqrt{36 \times 2025} \\ &= \frac{1}{6} + \frac{6 \times 45}{18} \\ &= 15 + \frac{1}{6} \end{aligned}$$

This suggests that $n = 15$ is possible. Checking, the number of rods when $n = 15$ is $3n(3n-1) = 45 \times 44 = 1980$ rods, which is possible. The next shape has $n = 16$ and requires $48 \times 47 > 45^2 = 2025$ rods, which is too many.

So the largest shape has 15 hexagons on each side. It requires 1980 rods and has $2000 - 1980 = 44$ rods left over,

hence (20).

30. Tim uses plug-in timers to automate his indoor plant lighting. Each timer has a plug on the back, a socket on the front, and a motorised dial that rotates once for every 24 hours that the plug has power.



When the time on the dial is between 9:00 and 17:00, the timer switch is on, so that any power on the plug is switched through to the socket.

For fun, he plugs together three of his timers, all set to time 0:00, with a lamp plugged into the front socket. How many hours until the lamp turns on?

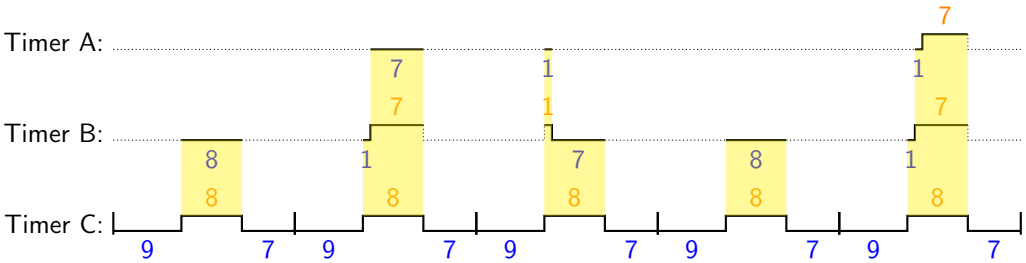
Call the timers A, B, C from front to back.

The lamp will turn on when timer A has received 9 hours of power, and is starting its 10th hour.

For this, timer B will have delivered 8 hours on its first rotation, then 1 hour on its second rotation. That is, it will have completed one rotation, and be showing 10:00. This means it has received 34 hours of power, and just be starting its 35th hour.

For timer C to deliver 34 hours of power, it will have completed 4 rotations, which will deliver 32 hours of power, and then another 2 hours between when it shows 9:00 and when it shows 11:00. Since C is receiving power continuously, this will be 4 days and 11 hours after the power was first switched on.

This diagram summarises what happens over 5 days:



The time it takes for the lamp to turn on is $4 \times 24 + 11 = 107$ hours,

hence (107).